
Chapter 2.1

APPLICATIONS OF SUMMARY MEASURES OF POPULATION HEALTH

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INTRODUCTION

Population health status has so many dimensions, classifications and measurements, as well as interactions between these, that patterns and trends often cannot easily be identified. Comparisons between populations and countries are often severely hampered because different data are collected or different classifications are used. Thus the need for comprehensive population health measures and standardization of data collection is obvious and longstanding. This paper describes the possible uses of such summary measures of population health (SMPH) and some conditions that have to be met for their application. But first, summary measures in general and their composition will be discussed briefly.

SUMMARY MEASURES

Summary measures combine information of different types and from different sources, which together permit a rapid appraisal of a specific phenomenon. They may be used for identifying differences (e.g. between groups, countries, etc.), observed changes over time, or expected changes (e.g. as a consequence of policy measures). There are four key elements in the construction of summary measures: the selection of relevant parameters to be included, the reliable measurement/collection of these parameters, the unit in which the summary measure will be expressed, and the relative weight of each of the constituents in the total summary measure. Each of these four elements will determine how the summary measure may be applied and for what kind of applications it will be valid. Ideally a summary measure has a wide range of valid applications, but inevitably there is a trade-off between the specificity of the information that is contained in the measure and its general applicability.

The value of summary measures themselves is in producing efficient and meaningful quick-glance information, generally for non-specialists. A World Bank economist wants the information behind a GNP, and a stock exchange (e.g. the Dow Jones Industrial Average) index has very limited value for a professional investor. Nevertheless, nobody doubts the usefulness of these indicators, because of their information value and the discipline that goes into the standardized collection of the constituent data.

ELEMENTS OF SUMMARY MEASURES FOR POPULATION HEALTH

If we define population health status as the amount and the distribution of morbidity, handicap and mortality in a population, it follows that the summary measure will have to include those three aspects of ill health. Mortality can be summarized in many ways, with probably the most important distinction between expectancy measures (life expectancies) versus health gap measures (years of life lost, YLL).

The (partly related) axes for a *taxonomy of SMPH* are at least:

1. Life table population vs. real population
2. Health expectancy vs. health gap
3. Generic vs. disease-specific epidemiological input
4. Types of values/valuation procedures for expressing disability/disease in time units
5. Incidence versus prevalence measures
6. Incorporation of other values.

Together these represent a large group of SMPH, which generally will have some family resemblance or formal relationship. It should be kept in mind that, for instance, in comparing burdens of disease between countries, starting with exactly the same input, different summary measures and choices made within summary measures may yield widely different comparisons and rank orderings.

As has often been stated, SMPH inevitably have a normative character, because the choice of constituent elements represents a selection and the weights relating these elements imply a judgement about the relative importance of each constituent element. Summarizing mortality data, for instance, often implies value judgements about a reasonable or realistic life expectancy.

As mentioned already, combining a large amount of health data into one summary measure will inevitably result in information loss. This will especially occur when, owing to preventive and therapeutic interventions, the trends in incidence, duration and mortality of a certain disease evolve in different directions. These may have very different consequences for

prevalence- versus incidence-based SMPH. Applications directed at estimating present health care needs would be more prevalence oriented, whereas applications directed at estimating the consequences of interventions would have to be more incidence oriented. But SMPH themselves are not intended for the detailed study of time trends in the incidence, duration and mortality of specific diseases.

DISABILITY WEIGHTS

After the publication of the Global Burden of Disease (GBD) study, the disability weights (linking mortality and disability data) and the social preference weights (age weights and discounting) have stirred much debate, which may even have obscured the paramount importance of GBD's epidemiological ambitions. Such weights may and perhaps even should be the source of a never-ending debate, and it is certainly very unlikely that there will ever be complete agreement about a universal set of disability weights for specific conditions. Fortunately, in the computation of SMPH sets of disability weights can easily be replaced. The reasons for doing so may be changes in judgements about the relative importance of specific diseases or disabilities, the amount of suffering involved, or the social consequences of specific diseases (depending on the perspective of the weighting procedure), or different policy applications of SMPH. Fortunately, all empirical research suggests that rank ordering of diseases from these perspectives is rather invariant between populations and cultures, apart from some specific types of health problems. If rank ordering is more or less stable, it is largely the size of the weight that will determine how much a specific SMPH will accentuate the importance of morbidity/disability relative to mortality.

VERSATILITY

The fact that SMPH inevitably have a limited information content and are not very sensitive to small changes in population health implies that, for many applications, continuous monitoring of SMPH may not be very informative. The frequency with which SMPH for specific applications should be reported is at least partly determined by the signal/noise ratio of that specific SMPH. It is well possible that in rich countries with high population health levels, the consequences of effective and efficient interventions will be so marginal that they will not easily be reflected in the SMPH. It is certainly possible to construct SMPH that would be sensitive to such relatively small changes due to specific interventions, but they may be less relevant for developing countries and vice versa. Thus, versatility is an important criterion in judging the usefulness of SMPH. Perhaps the most important contribution arising from the development of SMPH is that the process sets an agenda for the selection of relevant population health parameters, their uniform classification, data collection, and the consistency checking of these data.

POSSIBLE APPLICATIONS OF SUMMARY MEASURES OF POPULATION HEALTH

The reasons for developing SMPH are that diseases may vary widely in duration, severity, and lethality and that diseases may vary over time with respect to these aspects. Thus, SMPH typically combine information on morbidity, disability and mortality, generally expressed in time units (life-years, life expectancy).

The paper by Murray, Salomon and Mathers (chapter 1.2) lists eight applications.

1. Comparing the health of one population to the health of another population.
2. Comparing the health of the same population at different points in time.
3. Identifying and quantifying overall health inequalities within populations.
4. Providing appropriate and balanced attention to the effects of non-fatal health outcomes on overall population health.
5. Informing debates on priorities for health service delivery and planning.
6. Informing debates on priorities for research and development in the health sector.
7. Improving professional training curricula in public health.
8. Analysing the benefits of health interventions for use in cost-effectiveness analyses.

Applications 1 to 3 can be considered as descriptive, whereas applications 4 to 6 can be considered as intervention- or policy-directed.

DESCRIPTIVE/EXPLANATORY APPLICATIONS

The descriptive uses (comparing the health of populations, identifying time-trends, and describing the distribution of health within populations) are generally not goals in themselves, but are intended to identify differences or changes that deserve explanation and further exploration. For instance, the observation that mortality decline in the Netherlands at both young and old ages is somewhat lagging behind those of other rich countries, sparks off both scientific research on the determinants of this phenomenon and public debate about the performance of the Dutch health care system.

In order to compare health status between populations and to estimate time-trends in health status, the paramount requirement is that measurements and weights should be standardized and stable between countries and over time. Standardized information implies identical classifications, measurement and representation of collected data. For comparisons be-

tween populations, the availability of data for each constituent element would be a second requirement. For time-trends, invariance of classifications, etc. over time is obviously crucial and at the same time difficult to achieve, owing to developing knowledge and insights, as is illustrated by the successive editions of the International Classification of Diseases (ICD). Considering the required invariance of weights, their advantage is that it is a matter of consensus which set of weights will be applied, and that the sets of weights can be replaced in order to compute SMPH for specific purposes. Weights that strongly represent social value choices are less likely to be invariant over time.

POLICY APPLICATIONS

SMPH are relevant for three levels of health policy. The first level, general socioeconomic policy, is not often seen as health policy, but as long as poverty is still the major predictor of health, it should be. The systematic presentation of the distribution of health within populations and between populations should form an important input for socioeconomic policy. The second, more specific, health policy level focuses on the elimination or reduction of specific diseases or risk factors. For the planning of programmes for health protection, health promotion and immunizations and for the evaluation of such policies, the incidence- and mortality-oriented SMPH will be particularly important and helpful. The third category of health policy is specifically directed to health services. SMPH directed at informing this type of policy-making will generally focus more on prevalence measures and use weights that draw more attention to the effects of non-fatal health outcomes.

For setting agendas in biomedical research, information on the relative magnitude of health problems is very important. Obviously, other considerations, such as the intrinsic dynamics of specific research areas and the likelihood of developing breakthroughs will be at least as important. Nevertheless, the perception of a specific population health problem, either because of its present magnitude or because it may be an important threat, may often form the impetus for highly successful research, as is illustrated by many cases of especially infectious disease research.

SOCIAL VALUE CHOICES

Social value choices are meant to highlight specific elements in the health distribution within a population. Mostly this applies to age, gender, socioeconomic status or ethnicity. Social value choices may also refer to specific patient groups and specific diseases. Some brief remarks are given here.

1. Social preferences may differ between countries and change over time within countries. The use of SMPH for comparisons of burden of disease between countries or for documenting changes over time suggests that the weights relating morbidity and mortality should be as epide-

miologically oriented as possible, and that the SMPH should at least be reported without applying overall social value distributions, such as age-weights.

2. An obvious problem is that any weight to be applied to disease stages or generic health states always implies a form of interpretation (suffering, social limitations, health care use, social preferences) which may be relevant for some applications but not for others.
3. Policy-makers generally do not like the policy choices being made for them, somewhere hidden in computations. They sometimes like the consequences of different policy options to be made explicit.
4. The theoretically most universally applicable SMPH would stick as much as possible to its epidemiological basis and illustrate the consequences of different social value choices in a transparent way.
5. As health policy generally is directed at specific health problems or risk factors, the relation between these and the summary measure should remain clear.

ECONOMIC EVALUATION

Economic evaluation has traditionally used quality-adjusted life years (QALYs) as a summary measure combining mortality and disability. It is conceivable that standard evaluation techniques used in health economics may be integrated with more population health-oriented SMPH. This would require an endeavour to bridge traditional disciplinary differences, specifically the discussion of whether interventions or diseases should be the focus of study. For health policy and health care applications of economic evaluation, disease label/diagnostic information is indispensable. Allocation decisions about advanced transplantation surgery versus long-term care for mentally handicapped will (hopefully) never be decided on rank ordering of cost-effectiveness ratios.

DISEASE-SPECIFIC VERSUS GENERIC HEALTH STATUS INFORMATION

There have been many debates on whether SMPH should be based on disease-specific information, or on generic health state information, such as activities of daily living and generic quality of life measures.

Disease labels/diagnostic categories contain a wealth of information (see chapter 9.1). Although some may have limited value in establishing present and future health status in individual patients, they effectively summarize the distributions of present and future health status (including survival) in populations. They also are a shorthand for possible changes in prognosis (survival and future health status) through preventive and therapeutic interventions. Because they represent large bodies of scientific knowledge of determinants and disease processes, they are naturally at the core of both quantitative and pathophysiological biomedical research. In short, diagnostic categories are very effective heuristics for research, interventions and

the understanding of suffering. On the other hand, generic health status descriptions are indispensable to put disease-specific information in perspective at the population level.

CONCLUSIONS AND RECOMMENDATIONS

1. The development of summary measures of population health is of utmost importance for many purposes. Perhaps the most important contribution in their development is not the end product, the summary measures themselves, but the fact that they force us to define exactly the required epidemiological input on mortality and disability, to stimulate the application of uniform classifications and the collection of missing information, and to systematically check the internal consistency of the epidemiological information on specific diseases. Thus, apart from setting health and research agendas they also set agendas for international comparative data collection and epidemiological research. In that respect, SMPH development will always remain work in progress, of which the periodical publication of a set of indicators is just one, albeit important and attractive, result. Another very important contribution in the development of SMPH is that it structures the debate on social preferences in health policy and that the consequences of different social preferences may be quantified in terms of population health status.
2. In the debate on SMPH, often the weights relating mortality and disability draw the most attention and generate the most discussion. This obscures the fact that the largest uncertainties surrounding SMPH stem from lack of or inconsistencies in epidemiological information. In the computation of SMPH, sets of disability weights can easily be replaced by other sets of disability weights, whereas the collection of needed epidemiological information and testing their consistency are major tasks.
3. SMPH inevitably have a normative character because their construction requires decisions on the relative weight of the epidemiological input. Nevertheless, in a taxonomy of SMPH we can include a continuum from more epidemiologically oriented SMPH to those that strongly include social preferences. For descriptive purposes the more epidemiologically oriented SMPH would be desirable, whereas for policy purposes the inclusion of social preferences may be justified.
4. Considering the previous conclusions, we must accept that there is no “one size fits all” solution for the development of SMPH. In order to keep the main purpose of the endeavour clear, to keep the discussions on the different elements focused, and to make them versatile instruments for the different purposes, a system of modular reporting is indispensable. This means that the basic epidemiological input that goes

into the construction of SMPH, including estimates to fill information gaps and corrections due to consistency checks, should always be available separate from sets of weights. Such a basic dataset should allow for the computation of a variety of SMPH for different applications. This is also important for comparisons between populations directed at specific groups of diseases or at distributions within populations, and for the future construction of time series, when different disability weights or social preferences might become relevant.

5. For all applications, SMPH based on diagnostic categories are more useful. Diagnostic categories represent a wealth of information that will direct the search for determinants and interventions and will support the setting of priorities in health care and research policy.